A New Sheath for Highly Curved Steerable Needles

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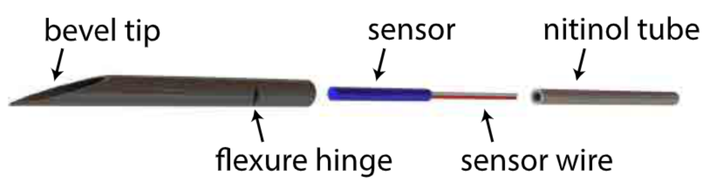
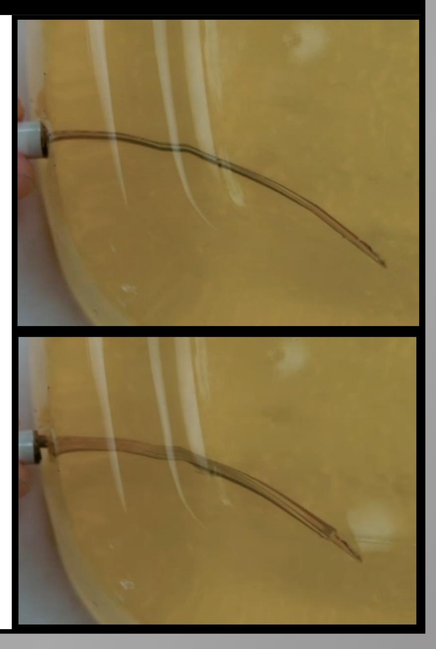
Steerable Needles

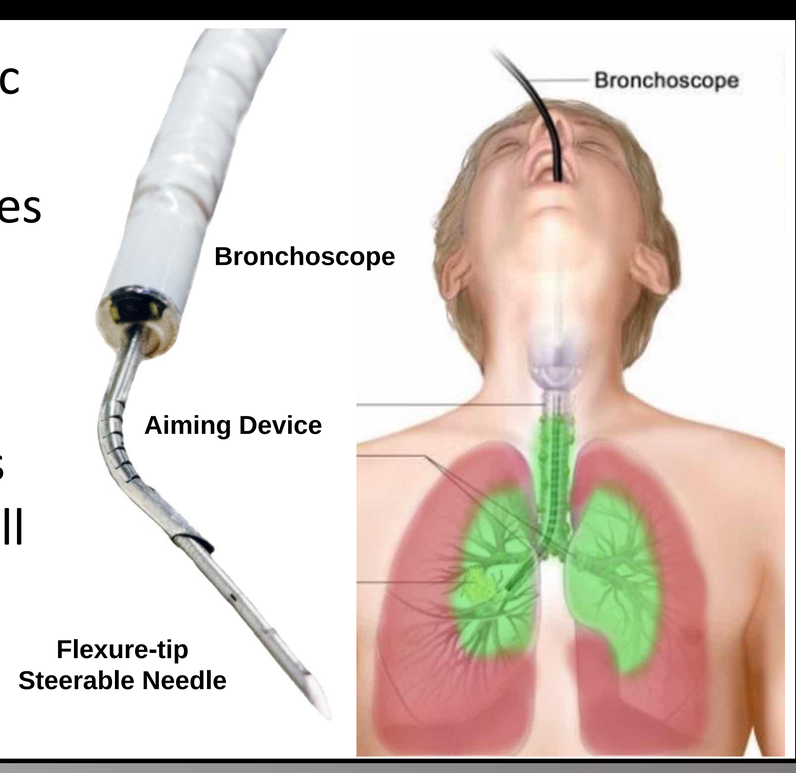
- Access hard-to-reach targets within the body
- Sheaths that slide over provide a channel for biopsy tools and therapy delivery



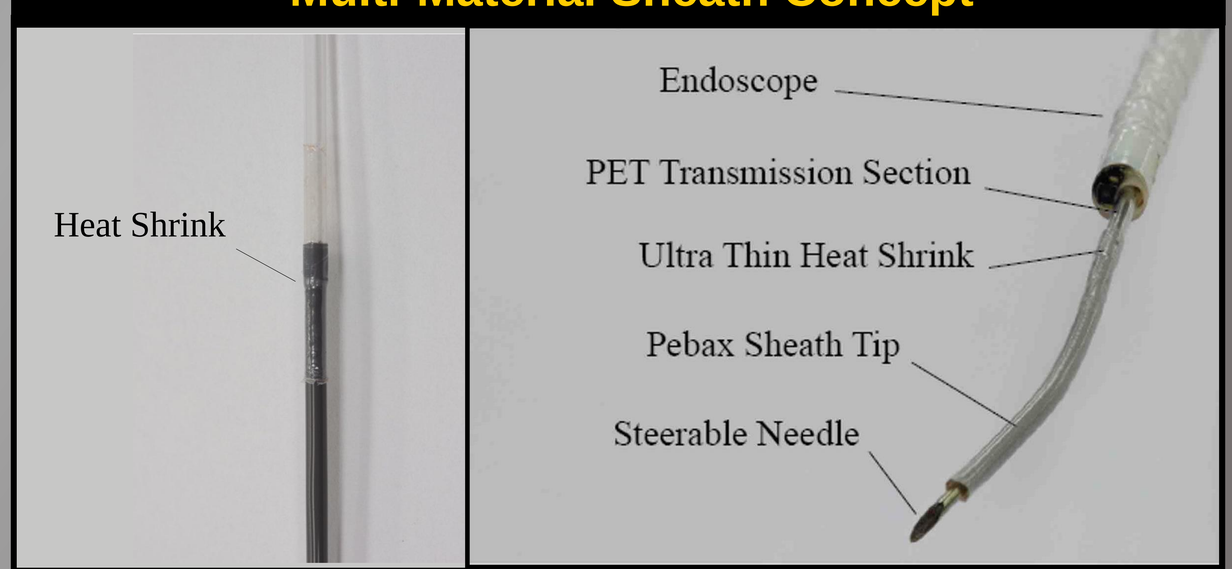


Problem

- Transendoscopic deployment of steerable needles requires longer needles and longer sheaths
- Existing sheaths do not work well due to friction, stiffness, etc.



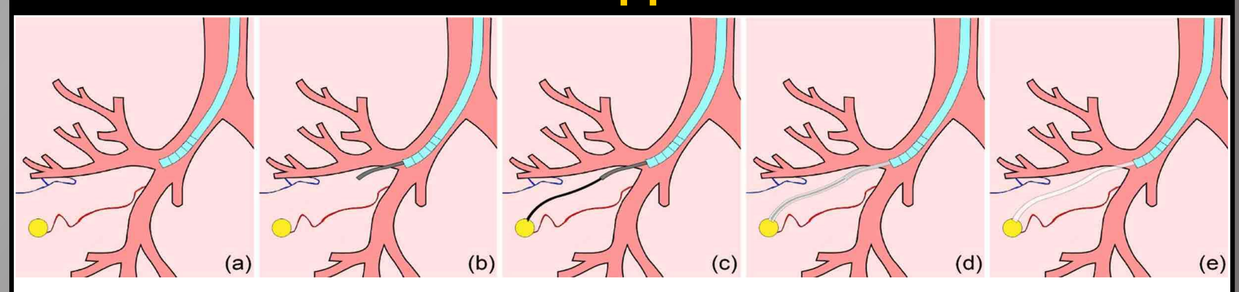
Multi-Material Sheath Concept



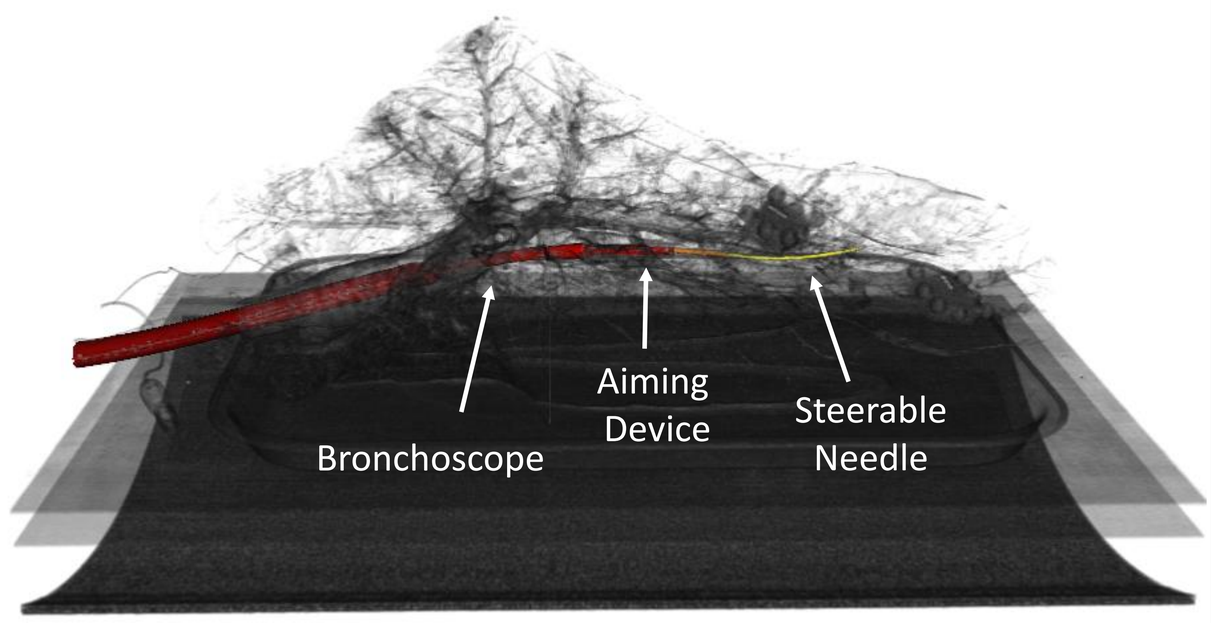
Two-Section Design:

- Transmission section: 1m high density polyethylene (PET)
  - Axially-Rigid and transmits motion well through endoscope
- Distal section: 150mm stainless steel braided Pebax 40D
  - Low bending stiffness

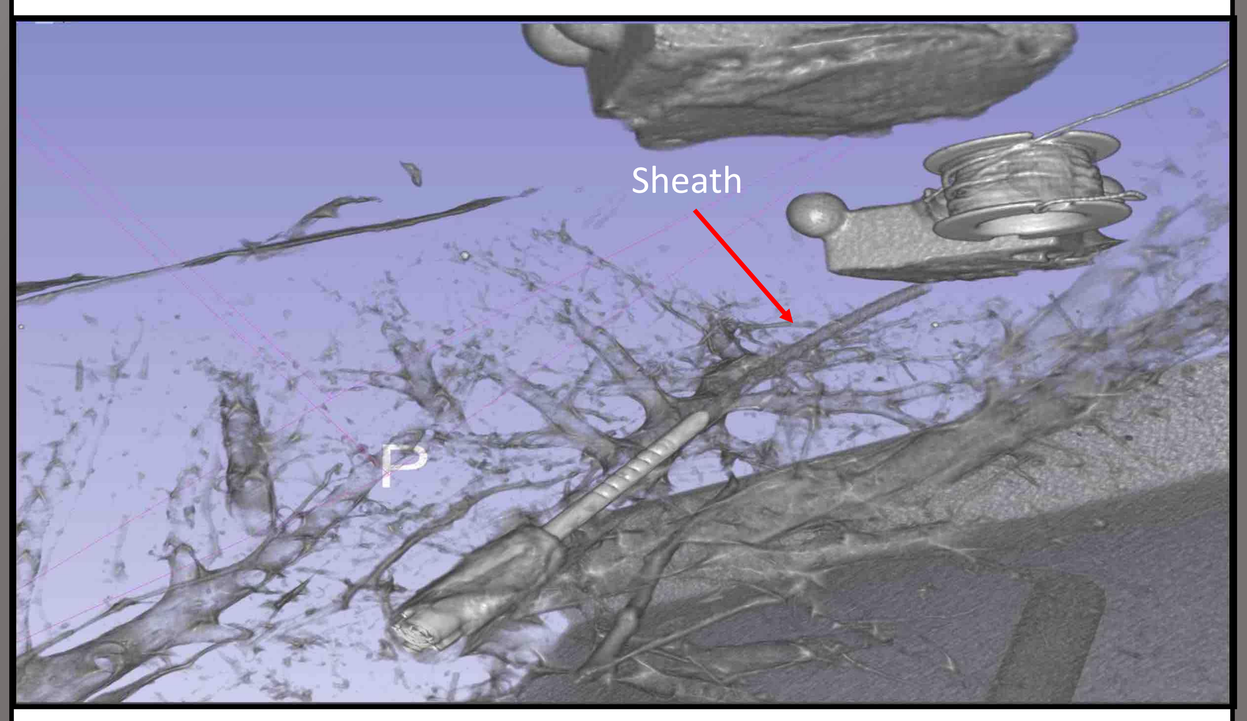
Clinical Application



Workflow: (a) Bronchoscope deployed (b) Puncture bronchial wall (c) Deploy steerable needle (d) Slide sheath over needle (e) Retract needle, leaving sheath



CT scan of *ex-vivo* inflated porcine lung with steerable needle system deployed to target.



CT scan of deployed sheath in *ex-vivo* inflated porcine lung

References

1. Majewicz, A., Marra, S. P., Van Vledder, M. G., Lin, M., Choti, M.A., Song, D. Y., and Okamura, A. M. (2012). "Behavior of tip-steerable needles in ex vivo and in vivo tissue", *IEEE Transactions on Biomedical Engineering*, 59(10), 2705–2715.
2. P. J. Swaney, A. W. Mahoney, B. I. Hartley, A. A. Remirez, E. P. Lamers, R. H. Feins, R. Alterovitz, and R. J. Webster, III, "Toward Transoral Peripheral Lung Access: Combining Continuum Robots and Steerable Needles", *Journal of Medical Robotics Research*, vol. 2, no. 1, p. 1750001, 2017.

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